


(Lecture starts from 12:15)

Q

```
function outer(){  
  var a = 10;  
  
  inner();  
  function inner(){  
    console.log(a)  
    var a = 30;  
    console.log(a)  
  }  
  var a = 1000;  
}  
console.log(a); → 11  
  
outer();
```

guess the o/p :- error : a not defined.

outer() fn call nahi hoga kyunki
l1 par error aa gya.

- ① GEC create hoga
- ② MCP mei outer fn ki definition aayegi.
- ③ Ab CEP mei l1 execute hoga, ab ~~cpu~~ computer dekhga ki a ~~keko~~ ki value kya hai inside MCP. It wont find anything there and current execution content ka koi parent nahi hai jaha jaaake a ki value dhundhi ja sake, isliye error aa jayega.

Q guess the o/p :-

```
function outer(){  
  var a = 10;  
  
  inner();  
  function inner(){  
    console.log(a)  
    var a = 30;  
    console.log(a)  
  }  
  var a = 1000;  
  
}
```

outer();

o/p : undefined
30

Q guess the o/p :-

```
function outer(){  
  var a = 10;  
  console.log(a);  
  
  inner();  
  
  function inner(){  
    console.log(a)  
    var a = 200;  
    console.log(a)  
  }  
  console.log(a);  
}  
console.log(a);  
  
outer();
```

o/p error a not
defined

Q

```
function outer(){  
  var a = 10;  
  console.log(a);  
  
  inner();  
  
  function inner(){  
    console.log(a)  
    var a = 200;  
    console.log(a)  
  }  
  console.log(a);  
}  
  
outer();
```

o/p → 10
undefined
200
10

Q

guess the o/p:-

10

10

```
function outer(){  
    var a = 10;  
    console.log(a);  
  
    function inner(){  
        console.log(a)  
        var a = 200;  
        console.log(a)  
    }  
    console.log(a);  
}  
  
outer();
```

Q

```
function outer(){  
    inner();  
    var a = 10;  
    console.log(a);  
  
    function inner(){  
        console.log(a)  
        var a = 200;  
        console.log(a)  
    }  
    console.log(a);  
}  
  
outer();
```

O/p $\Rightarrow$ undefined
200
10
10

Q

```
function outer(){  
    var a = 10;  
    console.log(a);  
  
    if(true){  
        console.log(a);  
        var a = 200;  
        console.log(a);  
    }  
    console.log(a);  
}  
  
outer();
```

$\Rightarrow$ o/p : 10
10
200
200

$\rightarrow$ yaha 200 isliye aayega kyunki
var ka scope yaha global hoga
Bcoz there is no function.

Q

```
function outer(){
  let a = 10;
  console.log(a);

  if(true){
    console.log(a);
    a = 200;
    console.log(a);
  }
  console.log(a);
}

outer();
```

O/p ⇒ 10
10
200
200

l3 wale block ki local memory mei a nahi hai isliye vo parent ke पास a check karega.

l4 par parent ki memory mei a = 200 ho gaya

l5 → a console pe 200.

l6 → l6 ki local memory mei a was changed to 200 from 10. 200 console.

isme wala code browser mei break points lagake solve kar naya tha.

Q

```
function outer(){
  let a = 10;
  console.log(a);

  if(true){
    console.log(a);
    let a = 200;
    console.log(a);
  }
  console.log(a);
}

outer();
```

O/p ⇒ 10

error

↳ cannot access a before initialization

l3 ke block ki local memory mei a declared hai isliye wo parent ka "a" consider nahi karega.

l3 par error aa jayega, then aage code nahi chalega.

agar l3 remove kar de tab l6 par o/p hoga $\rightarrow 10$

isme wala code browser mei break points lagake solve kar naya tha.

①

```
function outer(){  
    var a = 10;  
    console.log(a);  
    inner();  
  
    function inner(){  
        var a = 200;  
        console.log(a)  
    }  
    console.log(a);  
}  
  
outer();
```

O/P $\Rightarrow$ 10
200
10

②

O/P $\Rightarrow$ 10
200
10

```
function outer(){  
    let a = 10;  
    console.log(a);  
    inner();  
  
    function inner(){  
        let a = 200;  
        console.log(a)  
    }  
    console.log(a);  
}  
  
outer();
```

Q o/p ⇒
10
200
10

```
function outer(){  
  
    let a = 10;  
    console.log(a);  
    inner();  
  
    function inner(){  
  
        var a = 200;  
        console.log(a);  
    }  
    console.log(a);  
}  
  
outer();
```

Q o/p ⇒

```
function outer(){  
  
    let a = 10;  
    console.log(a);  
  
    {  
        let a = 200;  
        console.log(a)  
    }  
    console.log(a);  
}  
  
outer();
```

o/p ⇒
10
200
10

Q

```
function outer(){  
  
    let a = 10;  
    console.log(a);  
  
    {  
        var a = 200;  
        console.log(a);  
    }  
    console.log(a);  
}  
  
outer();
```

⇒ o/p ⇒ error

→ iss code mei var a = 200 ko
var b = 200 karke
browser mei break point banana,
tab jada aache se
samajh aayega.

let a & var a dono ko functionalscope hai, isliye
error aaye kyunki let a ko aap redeclare nahi kar
sakte same scope mei.

First Code Explanation

javascript

Copy code

```
function outer() {  
  let a = 10; // Declares `a` in the function scope  
  console.log(a); // Prints: 10  
  
  {  
    let a = 200; // Declares a new `a` in block scope (this does not affect the  
    console.log(a); // Prints: 200  
  }  
  
  console.log(a); // Prints: 10 (the outer `a` remains unaffected)  
}  
outer();
```

- The `let` keyword is **block-scoped**, so the inner `let a = 200` is confined to the block `{ ... }`.
- The outer `a` remains unchanged outside the block, and there's no conflict.

Second Code Explanation

javascript

Copy code

```
function outer() {  
  let a = 10; // Declares `a` in the function scope  
  console.log(a); // Prints: 10  
  
  {  
    var a = 200; // Error: Identifier 'a' has already been declared  
    console.log(a);  
  }  
  
  console.log(a);  
}  
outer();
```

- The `var` keyword is **function-scoped**, so it tries to declare `a` in the same function scope where `let a` is already declared.
- This causes a **SyntaxError: Identifier 'a' has already been declared**, as `let` does not allow redeclaration of the same variable in the same scope.

(1:26:00)

HOF (High order functions) :-

```
function sum ( num1 , num2 ) {  
    console.log ( num1 + num2 );  
}  
sum ( true , false );
```

o/p $\rightarrow 1+0 = 1$ (type conversion ho gya)

```
function a ( fn ) {  
    console.log ( "mai hu a" );  
    fn ();  
}  
  
function b () {  
    console.log ( "mai hu b" );  
}  
  
a ( b );
```

function b is passed as an argument.

$fn = b \Rightarrow fn$ is a parameter

agar function ke argument mei koi dusra function as an argument pass kiya gya hai, then aise function ko high order function.

a is HOF.

b is call back function.

A higher order function is a function that takes one or more functions as arguments, or returns a function as its result.

```
function a(fn){  
  console.log('mai hu a');  
  fn();  
}  
  
function b(){  
  console.log('mai hu b');  
}  
  
a(b);
```

```
function a(fn){  
  console.log('mai hu a');  
  fn();  
}  
  
a(function b(){  
  console.log('mai hu b');  
});
```



Both are same codes

(1836:00)

yaha hamne function b() ko
poora uthake a ke brackets mei
pass kar diya.

20 Nov, 2024

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```
function a(){
  console.log("inside a");

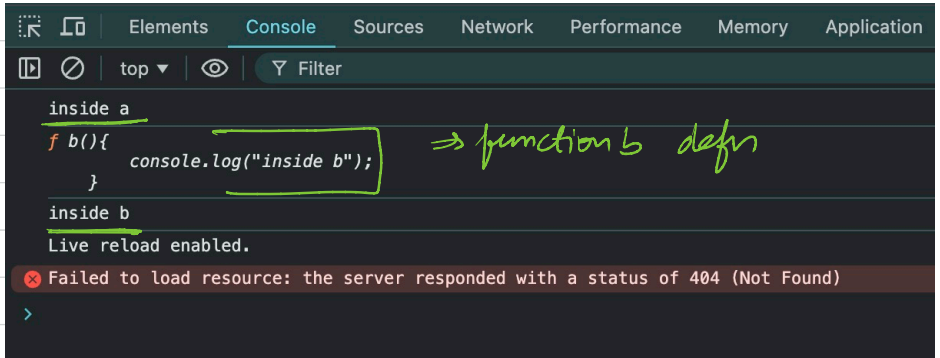
  function b(){
    console.log("inside b");
  }

  return b;

  let temp = a();
  console.log(temp);
  temp();
}
```

func a is HOF.

func b is call back function.



yaha function a is HOF kyuki vo function b return kar raha hai.

call back function:-

```
function a(fn){
  console.log('mai hu a');
  fn();
}

function b(){
  console.log('mai hu b');
}

a(b);
```

→ func b is a callback function because b is passed as argument inside function a. Also function b is called inside function a.

```
function a(fn){
  console.log('mai hu a');
}
```

? → fn() is missing

```
function b(){
  console.log('mai hu b');
}

a(b);
```

⇒ fnc b is not a callback function kyuki uss function ko function a ke andar call nahi kiya gya
fn() is missing.

the function which is being sent as an argument to some other function and therein called as well then it is a CALLBACK function.

↳ Samarth sir definition

A callback function is a function that is passed as an argument to another function and is called within that function, often to complete some operation or handle a specific event

↳ according to gpt

A callback function is a function passed into another function as an argument, which is then invoked inside the outer function to complete some kind of routine or action.

Q Kya function b call back hai?

fnc b is not cb fndtion.

But function a is HOF.

```
function a(){
  console.log("inside a");

  function b(){
    console.log("inside b");
  }

  return b;
}

let temp = a();
console.log(temp);
temp();
```


Since b is not passed as an argument to another function, it doesn't qualify as a callback function in this case

Note:- call nahi hua function, matlab call back function nahi hai.

Q Write a program to ~~get~~ get either all strings / all numbers / all integers from an array.

let arr = ['sam', 100, 'police', true];

```
let arr = ['sam', 100, 'bhaukaal', true, 400, false];
```

```
function getString(arr){
```

```
  let res = [];
```

```
  for(let item of arr){
    if(typeof(item)=== 'string'){
      res.push(item);
    }
  }
```

```
  return res;
}
```

```
function getNumber(arr){
```

```
  let res = [];
```

```
  for(let item of arr){
    if(typeof(item)=== 'number'){
      res.push(item);
    }
  }
```

```
  return res;
}
```

```
function getBool(arr){
```

```
  let res = [];
```

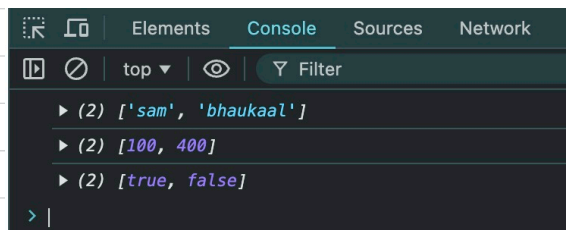
```
  for(let item of arr){
    if(typeof(item)=== 'boolean'){
      res.push(item);
    }
  }
```

```
  return res;
}
```

```
console.log(getString(arr));
console.log(getNumber(arr));
console.log(getBool(arr));
```

yaha iss code mei hamne almost same function 3 baar likha hai. Not efficient. We will use CB & HOF.

⇒ o/p ↴



```
Elements Console Sources Network
top Filter
▶ (2) ['sam', 'bhaukaal']
▶ (2) [100, 400]
▶ (2) [true, false]
> |
```

we will reduce the code_n using ^{length} HOF:-

```
function getString(item){
    return typeof(item)=== 'string'
}
function getNumber(item){
    return typeof(item)=== 'number'
}
function getBoolean(item){
    return typeof(item)=== 'boolean'
}
```

```
function get(arr , fn){
  let res = [];
  for(let item of arr){
    if(fn(item)){
      res.push(item);
    }
  }
  return res;
}
```

→ main driver
code

```
let arr = ['sam', 100, 'bhaukaal', true, 400, false]
```

```
console.log(get(arr , getString));  
console.log(get(arr , getNumber));  
console.log(get(arr , getBoolean));
```

1st samajhna jaruri hai $\Rightarrow$ item = arr[0], arr[1], etc

if (true) {
 }
} → fn (item)

agar fn (item) true hua then value array
mei push kar do.

get() → HOF

cb → getString()

